

2	C	Horizontally, $u \cos \theta \times t = s_x \rightarrow u \cos \theta = \frac{100}{2.0} = 50 \quad \text{--- (1)}$ Vertically, taking upwards as positive, $s_y = u \sin \theta \times t - \frac{1}{2} a_y t^2 = 0$ $u \sin \theta = \frac{1}{2} (9.81)(2.0) = 9.81 \quad \text{--- (2)}$ $\frac{(2)}{(1)}: \quad \tan \theta = \frac{9.81}{50} \rightarrow \theta = 11^\circ$
3	B	By relative speed of approach = relative speed of separation